

Revised Generation Interconnection System Impact Study Report

For Queue Project Q21-1827

Mccabe 138 kV
April 29, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1827. The Interconnection Customer proposes a 89.6 MW battery energy storage facility interconnecting at the Mccabe 138 kV substation in the North Carolina area. The request was received on July 23, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 491 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 13 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$17,721,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Trowbridge 138/69 kV Auto T1	Bel Air to Fabrica Tenere 138.0 kV	146.1	92	24.1
Jajome 343 138/69 kV TX Unit 1	Jajome 3403 138 kV/69 kV TX Unit 2	145.6	145	61.5
Trowbridge 138 kV/69 kV Transformer Bank	Bel Air to Fabrica Tenere 138.0 kV	140.7	96	24.1
Jajome 3403 138/69 kV Transformer Bank	Jajome 3403 138 kV/69 kV Auto	128.8	164	61.5
No Maries to Trowbridge 138.0 kV	Bel Air to Fabrica Tenere 138.0 kV	124.9	235	48.2
Bel Ir to Fabrica Tenere 138.0 kV	Home Substation to No Maries 138.0 kV	123.2	249	50.9
No Maries to Home 138.0 kV L	Bel Air to Fabrica Tenere 138.0 kV	116.6	300	48.8
Home Substation to Bel Air 138.0 kV	No Maries to Home 138.0 kV	107.8	258	50.9
Wheelabrator Portsmouth to Gosport 138.0 kV	Avra Vly to Chuckatuck 138.0 kV	105.9	371	6.8
Cedar Heights 138/69kV Autotransformer	Laurel City 138/1kV Transformer	101.7	89	24.4
Cedar Hts 138 kV/69 kV Auto #2	Laurel City 138-1 kV Transformer Bank	101.7	89	24.4
Cedar Hts 138-69 kV Auto #3	Laurel City 138 kV/1 kV Auto	101.7	89	24.4
Jajome 3403 to Calistoga 199 Substation 138.0 kV	Mira Vista to Yarnen 345.0 kV	100.3	214	68.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Trowbridge 138-69 kV XFMR	Replace transformer with higher-rated unit	\$3,649,000	16% (\$594,000)
Jajome 3403 138 kV/69 kV Transformer Bank T1	Replace transformer with higher-rated unit	\$3,308,000	16% (\$523,000)
Trowbridge 138/69 kV XFMR Unit 2	Replace transformer with higher-rated unit	\$5,388,000	16% (\$876,000)
Jajome 3403 138 kV/69 kV TX	Replace transformer with higher-rated unit	\$5,708,000	16% (\$903,000)
No Maries to Trowbridge 138.0 kV	Rebuild line with higher-capacity conductor	\$13,929,000	16% (\$2,265,000)
Bel Air to Fabrica Tenere 138.0 kV L	Rebuild line with higher-capacity conductor	\$14,645,000	16% (\$2,332,000)
Home to No Maries Substation 138.0 kV	Rebuild line with higher-capacity conductor	\$17,089,000	16% (\$2,735,000)
Home Substation to Bel Air 138.0 kV	Reconductor line	\$13,499,000	18% (\$2,360,000)
Wheelabrator Portsmouth to Gosport 138.0 kV	Reconductor line	\$922,000	2% (\$22,000)
Cedar Heights 138/69kV Autotransformer	Replace transformer with higher-rated unit	\$5,441,000	18% (\$992,000)
Cedar Hts 138 kV/69 kV Auto #2	Replace transformer with higher-rated unit	\$5,594,000	18% (\$1,020,000)
Cedar Heights 138 kV/69 kV XFMR	Replace transformer with higher-rated unit	\$6,470,000	18% (\$1,180,000)
Calistoga 199 to Jajome 3403 138.0 kV	Reconductor line	\$12,154,000	16% (\$1,919,000)
	Total allocated to project		\$17,721,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.